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The Impact of Digital Technologies on Neoclassical Labour Market

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Abstract

Labour market economics works on the principle of a neoclassical approach. Competitive forces, corporate policy, collective bargaining, and state intervention create the reality of the market. Technological progress in advanced technology, robotics, and artificial intelligence is strengthening the economies of several countries. The article deals with the actual influence of digital technologies on the neoclassical paradigm of the labour market and the impacts on the tasks and requirements for the competencies of workers and the level of their qualifications. Research is based on secondary data from relevant sources with emphasis on the quality using the interface Primo database, enabling the search for data in full-fledged databases (Web of Science, ProQuest, etc.). The real minimum wage, employment rate and jobs at risk of automation were analysed to compare the situation in European countries. The main objective is to analyse the magnitude of the impact of digital technologies on the labour market. The results point to technological progress, especially automation, which eliminates job positions by substituting low-skilled workers with machine learning robots. Government can ensure more balanced inclusive growth in the area of education. The change in the concept of teaching leads to higher qualifications and skills for future employees.

Keywords

Neoclassical Economics, Labour Market, Digitalization, Digital Technologies

I. Introduction

Since the industrial revolutions, fears of job loss due to technology penetration have grown. In the 1920s, the American Republican Party decided to support specific industries that were not competitive due to lower foreign wages and low living costs abroad (Goldberg, 1962). Keynes in the 1930s pointed to the threat of technological unemployment. The threat of automation grew, and in 1961 US President Kennedy established the Office of Automation and Workforce. The Office's mission was to maintain full employment despite

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labour substitution. Despite these concerns, employment has grown in many countries and new jobs have been created for women (OECD, 2020).

In general, the growing trend of job creation continues, and the intensity of technological progress is increasing. Technological progress and globalization are key factors in strengthening the economies of many countries. The fourth industrial revolution is characterized by increased productivity, optimization of processes and production capacities, and automation of control systems of production processes. Dynamic development increases the requirements for cyber security and the resulting confidentiality and availability of processed data. Nevertheless, digitalization encourages innovation and generates new opportunities for companies. The implementation of digital technologies is a necessary condition for inclusive growth and strengthening the competitiveness of companies, and therefore digital technologies mitigate economic losses in case of an unprecedented situation (OECD, 2016). Digitalization is expected to lead to positive employment growth in occupational profiles associated with products or services benefiting from increased demand due to the adoption of new technologies affected by changed processes. Despite the growing benefits of digital transformation, potential threats to technological change are emerging. The public is concerned about the impact of technology on jobs. New requirements are emerging for the qualification of employees, and the competencies of employees are changing. There are growing concerns about the elimination of jobs due to low levels of skills and their substitution by industrial robots. Machine-based robots replace the lowest-skilled workers (Perret, 2017).

The article considers a neoclassical model that shows the dynamics of the labour market. The precondition for equilibrium is the pareto-optimal allocation based on setting the price at the level of marginal costs and the distribution of income among consumers. The interaction of labour supply and demand determines a certain level of employment and unit wages. The global manufacturing industry is facing major challenges due to technological change, consumer trends, and the skills requirements for workers changing. Specialized tasks require higher qualifications of employees. Changes related to digital transformation can affect the overall economic system.

The article discusses the magnitude of the impact of digital technologies on the neoclassical labour market, analyses the impact on employment, and identifies key threats and benefits. The work contains three chapters. The first part is an introduction. The following chapter deals with a literary research of the terms of neoclassical and classical approach of modern economics. In the next part of the literature review, the article deals with the interpretation of the concept of digitalization, which is an important part of the fourth industrial revolution. It was based on articles published in the Scopus and Web of Science databases. The third part is a discussion that summarizes the impact of digitalization on the neoclassical labour market and is supplemented by data from Eurostat and the OECD. Concluding remarks follow. The result is a summary of theories that focus on the interconnection of these two areas.

II. Literature Review

Neoclassical Approach

The neoclassical approach is based on the assumptions of satisfying a certain level of consumption of an individual, striving for individuals to maximize benefits, and striving for organizations to maximize profits. The central paradigm of neoclassical theory is marginalism. It is based on the theory of marginal value of utility (value theory) or loss resulting from changes in supply and demand, which are the basis for decision-making of individual entities in competitive markets, and the theory of marginal productivity, distribution theory (Sovani, 1966). The neoclassical approach derives the demand for labour from marginal productivity as income from the marginal product of labour, which is measured against the marginal cost per factor (Fanti and Manfredi, 2007).

Representatives of the neoclassical approach sought the laws of functioning of markets and entities. The forerunner of marginalism was the German economist Hermann Heinrich Gossen, who was followed by neoclassical officials. The main representative of neoclassical economics was Marshall, whose labour market model market demand is derived from a declining marginal product of labour (Marshall, 2013). Other representatives of this approach were Jevons, Walras, Barnett, and Morse. Neoclassical economics follows the basics of classical economics. It abandons the labour theory of values and moves to a new understanding of values as a reflection of preferences and production costs (Lisý, 2018).

The representative of classical economics was Adam Smith. He advocated the theory of the invisible hand of the market as a mechanism for the functioning of a market economy that effectively distributes income and production based on the free choice of individuals to pursue their own interests. The size of marginal products is a decisive factor in the distribution of labour. It is essential to determine the level of the wage rate and the allocation of labour. This theory has already experienced many restrictions in the form of government regulations, corporate regulations, and trade union negotiations, determining the situation on the labour market (Lisý, 2018). A number of other models of economic equilibrium are derived from Smith's theory, in which personal interests in a competitive environment bring the greatest benefit to society. Vilfredo Pareto used the social optimum to graphically display the so-called "Edgeworth box" diagram (Brožová, 2016).

In the book of Hicks (1963), the theory of marginal productivity defined the concept of substitution and deals with the demand for labour, the competitive market, and the restrictive measures of state regulations. Hicks was influenced by Robbins, who specified the conditions of a positively or negatively inclined individual job offer. For the first time, the article deals with a backward curved curve. The shape of the individual supply curve after work captures the income and substitution effect (Robbins, 1930).

The neoclassical Cobb-Douglas production function and the utility function represent a fundamental building block in the labour market. The two-factor production function represents the relationship between the number of inputs (capital, labour) and the size of the output (level of production). The elasticity of capital and labour substitution is a key parameter in analyses of the size of a potential product. Values are the starting point for

understanding long-term economic growth and its sustainability (Solow, 1956). A higher value of substitution elasticity represents a higher per capita income (Klump and Grandville, 2000). Elasticity also reflects the effects of fiscal and monetary policy. In models that consider a higher value of elasticity, changes in monetary policy have a higher impact on inflation (Erceg and Levin, 2006).

Chatterjee and Turnovsky (2007) dealt with the analysis of distributional dynamics in the neoclassical growth model and the role of flexible labour supply. From the result of the work, it is clear that in the absence of fixed costs or imperfections in the capital market, the accumulation of capital leads to a reduction in the rate of wealth inequality. The dynamics of income inequality is determined by three factors: the dynamics of wealth inequality, production factors, and leisure preferences. The change in leisure-time preference, which corresponds to structural changes, has a decisive impact on the flexibility of the labour supply in the distribution of income over time.

In a related article by Fanti and Manfredi (2005), it was found that robust cyclical and chaotic behaviour of prices and quantities occurs in the case of low substitution between consumption and leisure.

The theory of rational choice is based on the assumption of the behaviour of individuals regardless of the preferences of others in economic models. The main representative is Becker, who abandons the theory of values and focuses on the possibilities of modern economic theory. In competitive markets, people seek to maximize the benefits with respect to the amount of personal capital. Becker's theory of household production discusses the family's efforts to maximize utility through the efficient allocation of time, income, and the purchase of goods and services (Kaufman and Hotchkiss, 2006).

Persistent unemployment conditioned the emergence of new theories. These reflect market imperfections, asymmetric information, and inelastic wages. Stigler (1961) deals with obtaining information about the job offer. The opportunity cost reflects the rational behaviour of individuals in obtaining job choice information. In the event of a sub-optimal job offer, the unemployed person continues the job search activity. The decline in labour market demand indicates a state of redundancy, not a decline in wages. Azariadis (1975) defined the theory of implicit wages in which employees prefer a long-term lower wage to a higher indefinite one.

The model of economic behaviour, Homo oeconomicus, expresses human motivation based on personal interests. The goal is maximum utility with optimal satisfaction of their needs with minimum resources (Kaufman and Hotchkiss, 2006).

Digitalization

Digital transformation is a technological process of constant change in organizations. Digitalization transforms business processes across all industries and focuses on connectivity supported by digital technologies. As a result of the changing global business environment, companies are striving to achieve significant flexibility, accelerate response times, and gain new business opportunities (Parida, 2018). Digital transformation is expected to have high annual growth and simplify and streamline established processes, ensure higher data security, and increase the company's competitiveness and profitability. The digitalization

strategy requires a thoughtful approach with a specific plan, a list of stakeholders, identified benefits, and potential complications. To ensure intelligent production, automation is necessary, which leads to the transition to advanced technologies. The synergy of production and information technologies makes great progress. Productivity increases both in service systems and in production. The digital revolution is often referred to as the Industry 4.0 concept (Ustundag and Cevikcan, 2017).

ICT technologies have made significant progress and moved beyond the ability and understanding of the individual. This development is referred to as the fourth industrial revolution, which is characterized by artificial intelligence, machine learning, innovation, and communication (Lovergine and Pellero, 2018). It follows the third industrial revolution, which began in the 1960s and is associated with the birth of mainframe computers (Chinoracký and Čorejová, 2019). Industry 4.0 represents continuous innovation, the implementation of cybernetic systems, and automation in order to increase the competitiveness of organizations across industries. Artificial intelligence with the ability to process computational operations in large volumes of data generates far more accurate predictions. Automation is considered a key component of digital transformation and will affect the labour market (Lovergine and Pellero, 2018).

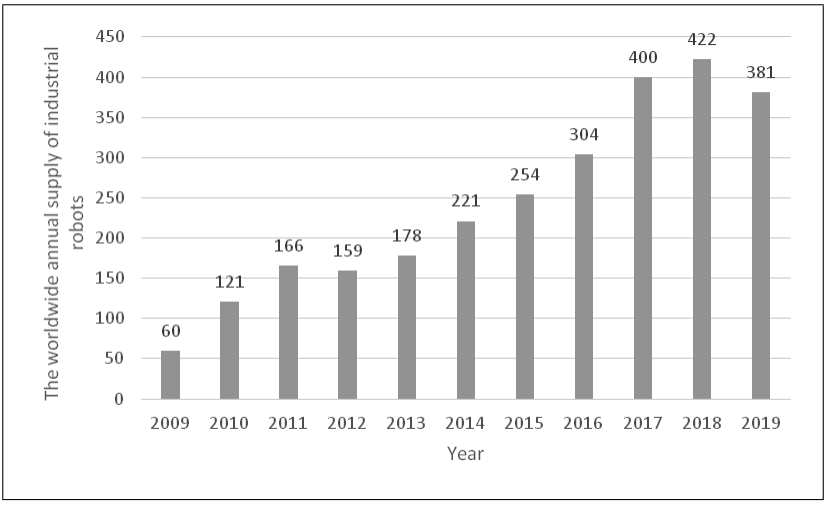
Kane et al. (2015) states that strategy, not technology, controls the process of digital transformation. The essence is the development of competencies in the use of digital technology. The transition to new technological trends within Industry 4.0 has a major impact on employment. The development of innovation, increased productivity, and reduced production costs has affected economic growth and achieved overall prosperity (OECD, 2017). Transformation brings serious employment challenges. As a result of rising wages, emerging markets are expected to replace individuals with average skills with robots. The interest of organizations in highly qualified workers is increasing. The demand for routine roles and tasks is decreasing; it is important to determine whether these tasks are cognitive or manual (Dachs, 2018).

Parida (2018) notes that as a result of business development and innovation management to address social and business values, the skills requirements of workers are changing significantly and jobs are becoming obsolete.

The global sales of industrial robots have tripled in recent years. The most significant increase in deliveries was recorded by the automotive industry. In countries where the automotive industry is highly represented, the prevalence of industrial robots is considerable (International Federation of Robotics, 2019).

The International Federation of Robotics (2019) published statistics on the annual supply of industrial robots in 2009–2019. A rapid increase in the supply of industrial robots is evident, and an analogous trend is expected. The trend of a growing share of national income in capital is fuelling a political debate on the concentration of capital ownership. Increased rate in use of robots and automation distributes revenue primarily to capital owners, not workers (Keeley, 2015).

Figure 1: Worldwide annual supply of industrial robots in thousands of units



Source: International Federation of Robotics (IFR, 2020)

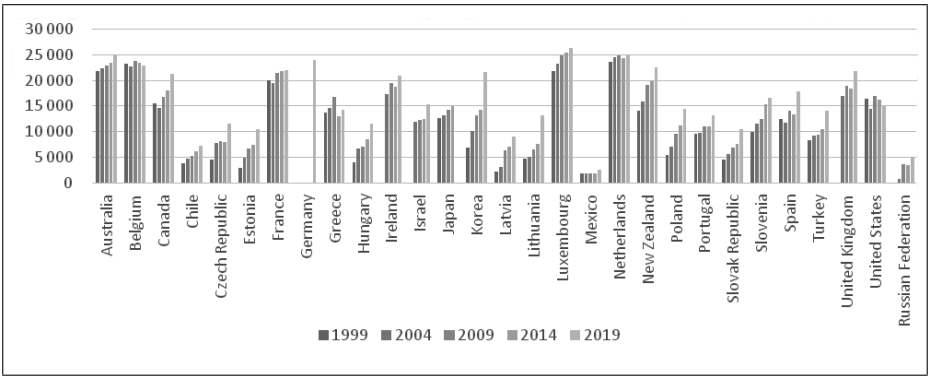
III. Results and Discussion

The Discussion chapter examines a connection between digitalization and the labour market. Globalization consists of a deeper integration of the production factors of individual states. The neoclassical approach is based on the assumption that the emergence of a given technology is conditioned by the substitution of production factors. Different levels of production are achieved by different combinations of capital and labour. Relative prices affect the amount of substitution. In the case of price changes in favour of the substitution of labour by capital, there is a transition from less capital-intensive technology to more capital-intensive technology (Nedomlelová and Kocourek, 2010).

An OECD study (2017) notes that new technologies and digitalization affect many aspects of life and have significantly transformed production processes by automating certain tasks. At the same time, transaction costs associated with communication and coordination around the world have been reduced, allowing for the vertical fragmentation of industrial production, which leverages the expertise and comparative advantages of different countries at different stages of production. The Italian economist Labini (1995), in the case of newly achieved values in the ratio of capital and labour prices, came to the conclusion that this eventuality would give rise to a new technology that would save labour. The transition to new technology is usually initiated by a request to increase output (production). The process of technological change takes time and is a dynamic problem of substituting inputs from production. Dynamic substitution is part of a long-term growth process, represented by mechanization, subsequent automation, and in many branches of robotization of production processes. This phenomenon reflects the relative rarity of work, which is reflected in rising real wages (Rodrigues, 2010).

The following chart shows the real minimum wages in the countries from 1999 to 2019. In the European Union, 22 countries currently have a legal minimum wage. Exceptions are Austria, Denmark, Italy, Sweden, Finland, and Cyprus (Adema et al., 2019). Comparisons between countries are difficult due to the pre-tax amounts indicated. The disposable income of households with the same minimum wage can differ significantly between the two countries. OECD statistics show that the highest rates of payroll tax are in the following countries: Belgium (52.24%), Germany (49.35%), Italy (48.01%), Austria (47.91%), France (46.67%), Hungary (44.58%), Czech Republic (43.94%). In contrast, the tax rate paid on wages within OECD countries is the lowest in the following countries: Chile (7.05%), New Zealand (18.77%), Mexico (20.14%), Switzerland (22.27%), Israel (22.72%). However, over the past five years, there has been a visible increase in the real minimum wage in most countries. The highest levels of the real minimum wage are largely achieved by EU countries (Belgium, France, Germany, Luxembourg, the Netherlands), as well as Australia, New Zealand, and Canada. Only the USA and Belgium recorded a real decline in the minimum wage.

Figure 2: The amount of the real minimum wage in USD in 1999, 2004, 2009, 2014, 2019

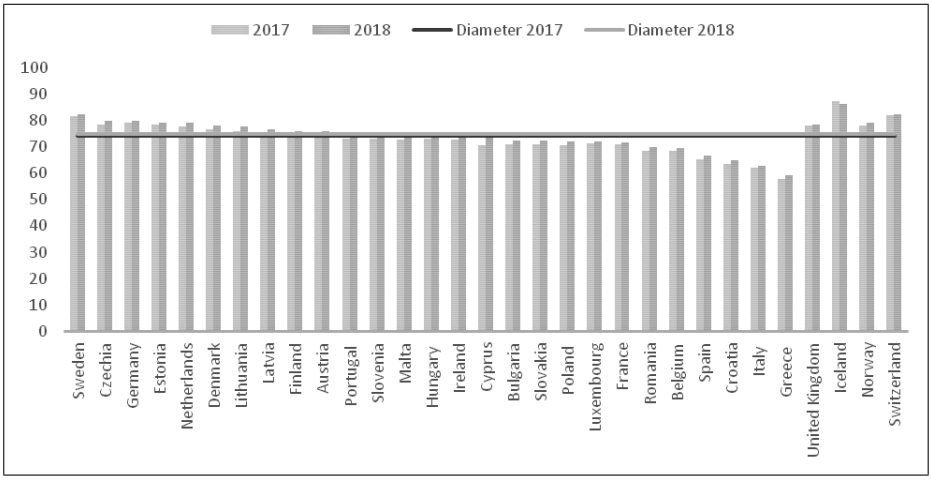


Source: OECD.Stat (2019)

In Figure 3, according to Eurostat, the employment rate of workers was 73.5% between 20 and 64 years of age in the European Economic Area in 2017. The following year, the overall rate increased to 74.73%. The employment rate increased in all countries except Iceland (−1.1 pp). The highest increase in employment occurred in the following countries: Cyprus (3.1 pp), Finland (2.1 pp), Greece (1.7 pp). The following countries recorded employment rates above 75%: Sweden (82.6%), Czechia and Germany (both 79.9%), Estonia (79.5%), the Netherlands (79.2%), Denmark (78.2%), Lithuania (77.8%), Latvia (76.8%), Finland (76.3%), Austria (76.2%), Portugal and Slovenia (both 75.4%). The lowest employment rate was recorded in Greece (59.5%), despite a year-on-year increase of 1.7 pp. This is followed by Italy with an employment rate of 63% (year-on-year growth of 0.7 pp), Croatia 65.2% (year-on-year growth of 1.6 pp) and Spain 67% (year-on-year growth of 1.5 pp).

Due to the growth of the employment rate, the real minimum wage is also rising. The theory of efficiency wages guarantees the payment of higher wages for which workers are willing to work. Low wages are demotivating and reduce labour effort and therefore productivity.

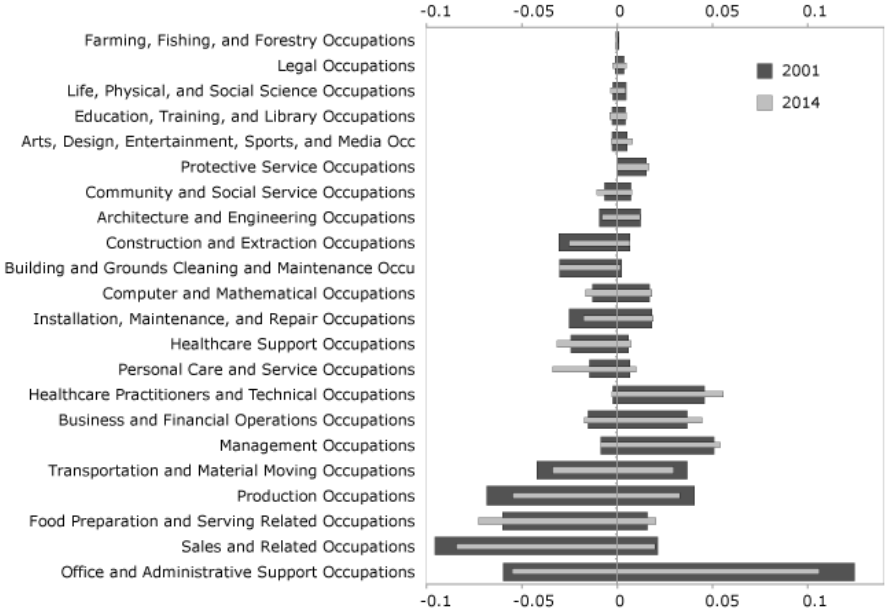
Figure 3: Employment rate by percent, 2017 and 2018



Source: Eurostat (2019)

When examining the growth of wages of specific professional groups, it is necessary to take into account the value of employment in a certain professional group. The following figure shows the individual professions, and the columns show the percentage of employment in the group. The green bars represent the values acquired in 2014, and the blue bars represent the year 2001. The value to the right of the axis indicates the share of the profession with wage growth and the value to the left indicates the share of the profession with wage decline. Occupations in the field of Personal Care and Service Occupations, Healthcare Support Occupations, and Food Preparation and Serving Related Occupations not only excel in a higher share of employment, but are occupations with declining real wages. Technical and health professionals employ more workers, but their real wages are higher in 2014 than in 2001. The professional groups discussed in the following show that most low-skilled jobs are valued at low wages.

Figure 4: Wage gains by percent of Employment, 2001 and 2014



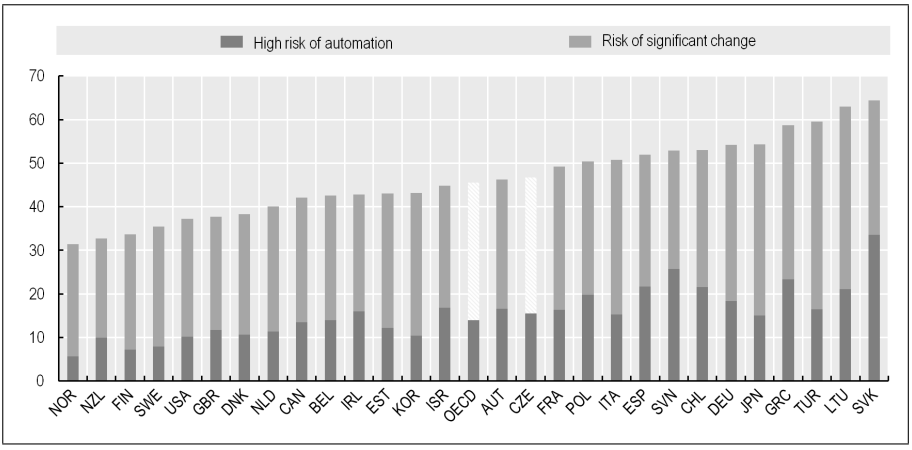
Source: DEED, Occupational Employment Statistics (2020)

OECD research (2017) noted a shift in employment from the manufacturing sector to the services sector as part of deindustrialisation. This process coincided with the polarization of the labour market, with the number of medium- and medium-skilled jobs declining compared to low- and high-wage jobs. The study by Lee and Clarke (2019) highlights the problems associated with the penetration of high technology. The gap between the wages of high-skilled workers and the wages of low-skilled workers is widening. Less skilled workers will be exposed to the risks of high housing costs and the risk of losing their jobs due to difficult qualification and skill requirements.

The OECD 2019 survey states that the share of jobs with a high risk of automation (the probability of automation is at least 70%) is on average 14% in the OECD. The values capture the potential elimination of jobs and do not take into account new jobs due to new technology. The analysis highlights the risk of automation for low-skilled workers.

The OECD membership consists of 30 member countries: Austria, Belgium, Canada, Denmark, France, Germany, Greece, Iceland, Ireland, Italy, Luxembourg, the Netherlands, Norway, Portugal, Spain, Sweden, Switzerland, Turkey, United Kingdom, USA, Japan, Finland, Australia, New Zealand, Mexico, Czech Republic, Hungary, Poland, Korea, Slovakia.

Figure 5: Jobs at risk of automation in OECD countries



Source: OECD (2019)

IV. Conclusion

The purpose of the paper was to analyse the magnitude of the impact of digital technologies on the labour market. The literature review is focused on the theoretical background, specifically the neoclassical approach and the concept of digitalization. The next chapter analyses the current situation and interprets the impact of digital technologies on the labour market. It was based on Eurostat statistics, research reports from the OECD, and the World Economic Forum. Innovative high-tech industries are a key factor in the growth of a country's economy. In recent years, there has been an increase in employment in the countries of the European Union. The average employment rate in 2018 is 74.73%. Despite the decline in 2019, the worldwide annual supply of robots is growing sharply. The level of the real minimum wage is increasing, although the gap between the real wages of higher-skilled and lower-skilled workers is widening. Due to technological progress, automation is a concern for low-skilled workers. Machine learning robots replace workers and eliminate jobs. This aspect lowers the living standards of individuals.

An increase in the employment rate of low-skilled working groups can be achieved by generating new jobs associated with the existence of new manufacturing companies. Government measures can ensure more balanced growth with the pursuit of prosperity with a broader social focus. An area that can ensure more inclusive growth is education. The government, in an effort to change the concept of teaching and motivate students from an early age, can influence the qualifications and skills of future employees (Keeley, 2015). The findings may lead to further reflection on improving the living standards of low-skilled workers. The subject of further research may be a deeper question: Who benefits from economic growth conditioned by digital technologies? How is capital-intensive production changing? How to close the gap between high-paid and low-paid workers? How to effectively shape the work environment and prevent possible risks due to robot penetration?

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